

Daniel McNab

UI & Graphics Engineer

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Cross platform software developer, with experience developing 2D Rendering and UI for Android. Proficient in Rust, with experience in TypeScript (React) and Kotlin. Open Source maintainer of Xilem, a UI library in Rust, and co-maintainer of Vello, a project consisting of several 2D renderers, specialised to different hardware.

Job Experience

UI & 2D Graphics Researcher - Contract February 2024 – Present

Google - *Vello and Xilem*

- Decreased startup time of Vello on Android by nearly two orders of magnitude, through contributions to wgpu.
- **Created the integration between the tracing Rust library and Android's native tracing API.**
- **Contributed API to wgpu to enable synchronised rendering with audio/visual content on Android.**
- **Created example demonstrating rendering using Rust code within existing Android apps.**
- Designed and implemented solutions for snapshot testing despite precision issues introduced by GPUs.
- **Ported the Vello Hybrid sparse strips renderer collaboration to Android.**
- Developed robust plain text input for Xilem.
- Developed a simple cross-platform Mastodon client as a hero example for Xilem.
- Reviewed contributions across the Linebender project as part of ongoing open source maintenance.

Web Developer - Full Time August 2023 – January 2024

PwC UK - Internal Website

- Led C# backend development for an internal web application, used for matching people with work suiting their skills.
- Developed a database design suitable for this application, and updated this as requirements changed.
- Supported developers working on the React single page application frontend, including providing accessibility guidance.
- Developed a solution for theme selection which live-updated across all local instances of the application.

Degree Apprentice - Full Time July 2021 – August 2022

PwC UK

- Completed a 14 month internship between 2nd and 3rd years of University.
- This was part of my level 6 Apprenticeship, with a Digital and Technology Solutions (Software Engineer) specialisation, achieved as part of my degree course.
- Supported a team working in UI automation by developing re-usable workflows (such as to navigate to a specific page in an internal app).
- Created processes to ensure routine report creation was accurate and efficient, unlocking more time for senior leadership.

Education

University of Leeds 2019 - 2023

BSc (Hons) in Computer Science (Digital & Technology Solutions) Leeds, England

Open Source Experience

linebender/xilem March 2024 – Present
Lead Maintainer

- Took over development of Xilem from Raph Levien, allowing him more time for fundamental 2D graphics research.
- **Ported several examples to Android.**
- Built the lifecycle integration with async operations and background tasks.
- Developed the second version of Xilem, which transitioned to a different widget toolkit with many more features.

linebender/vello October 2022 – Present
Maintainer

- Added support for COLR fonts, enabling Emoji support.
- Focused on developer experience, including Snapshot Testing and shader preprocessing.
- Collaborated with upstream projects (wgpu) as part of resolving issues.
- Provided review and design input as the Sparse Strips collaboration has developed.

bevyengine/bevy August 2020 – July 2022
Contributor

- Drove a relicensing effort to make Bevy more compatible with the rest of the Rust ecosystem.
- Significantly reduced noise when creating systems; each system previously needed a call to the `.system()` method.

Skills

Rust • Vulkan • WebAssembly (wasm) • WebGPU • WGSL • Kotlin • TypeScript • WebGL2 • C# • SQL • Tailwind

Tools

GitHub • VSCode • git • Tracy • Renderdoc • DaVinci Resolve